

Module 3 Unsupervised Learning – Clustering

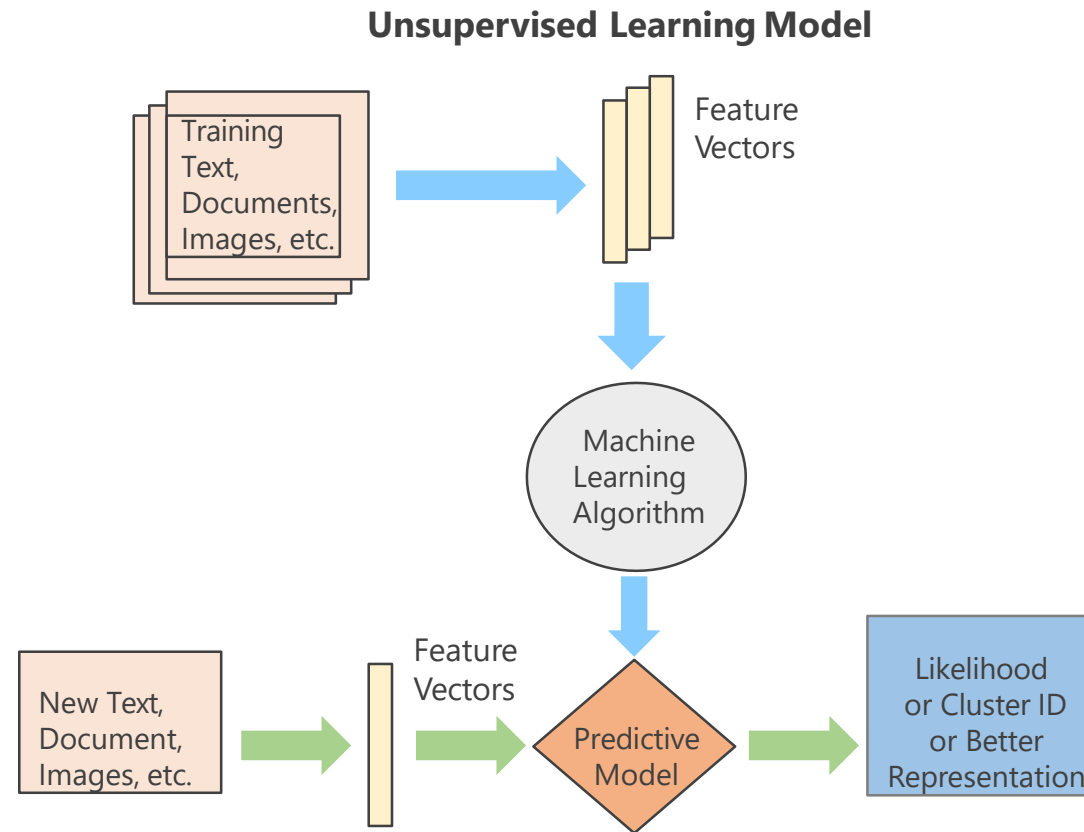
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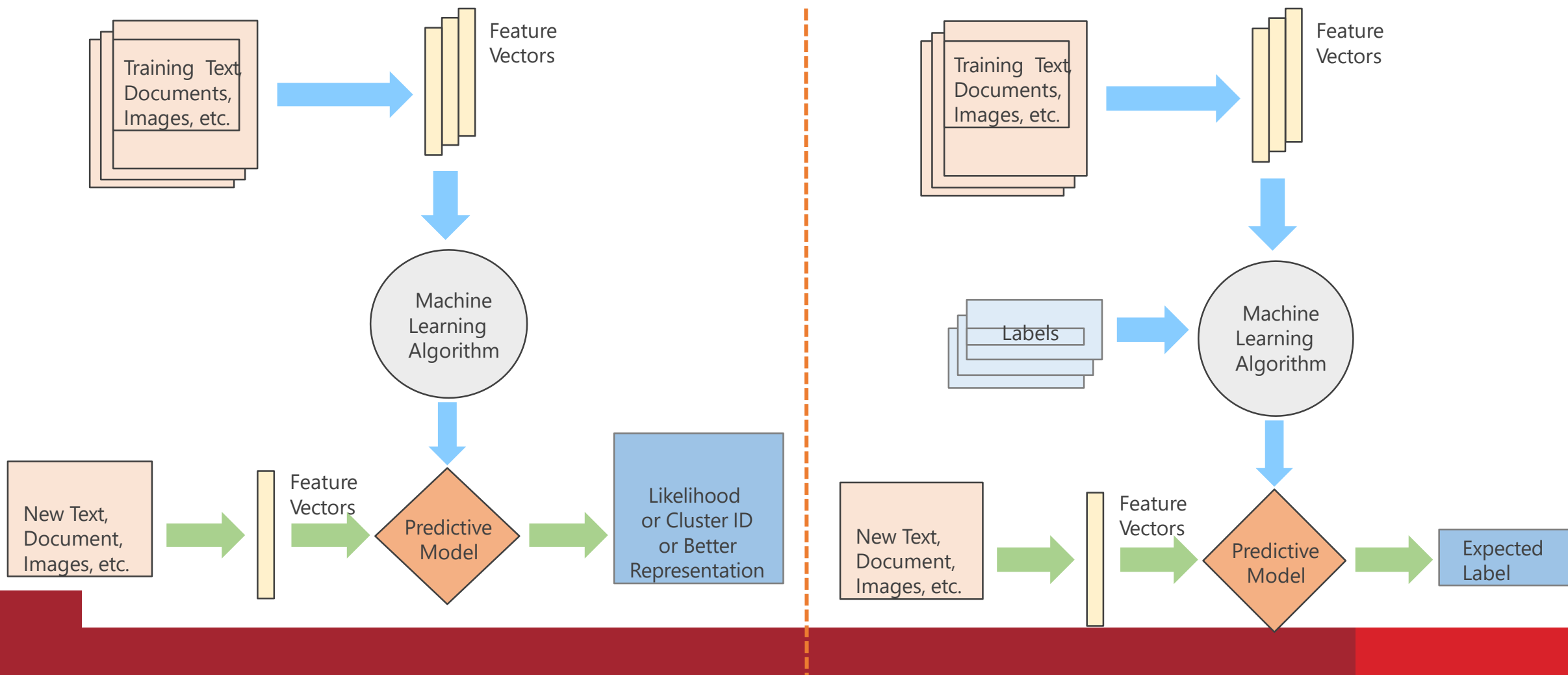
Unsupervised Learning Process Flow

The data has no labels. The machine just looks for whatever patterns it can find.



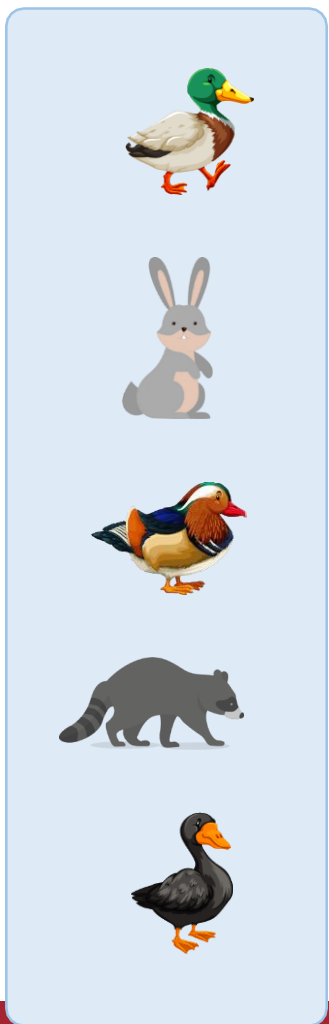
Unsupervised Learning vs. Supervised Learning

The only difference is the labels in the training data

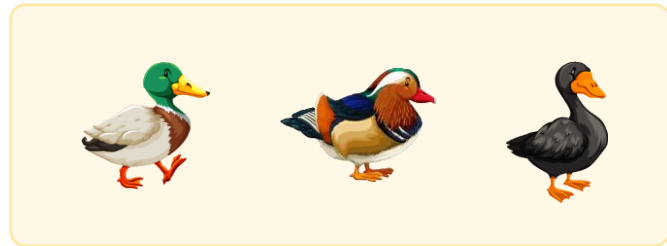


Unsupervised Learning: Example

Clustering like-looking birds/animals based on their features

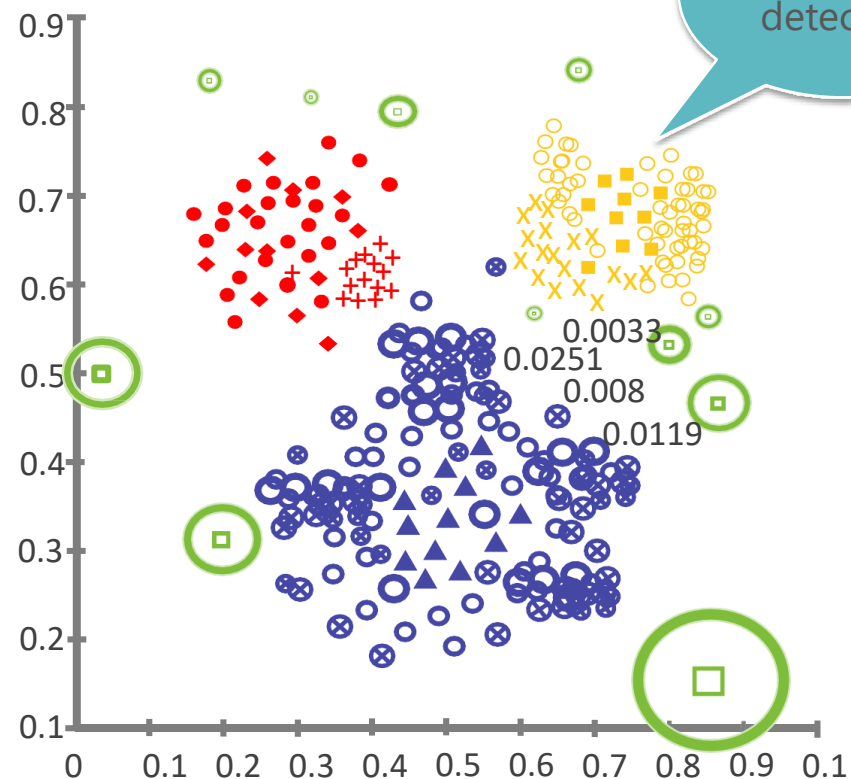
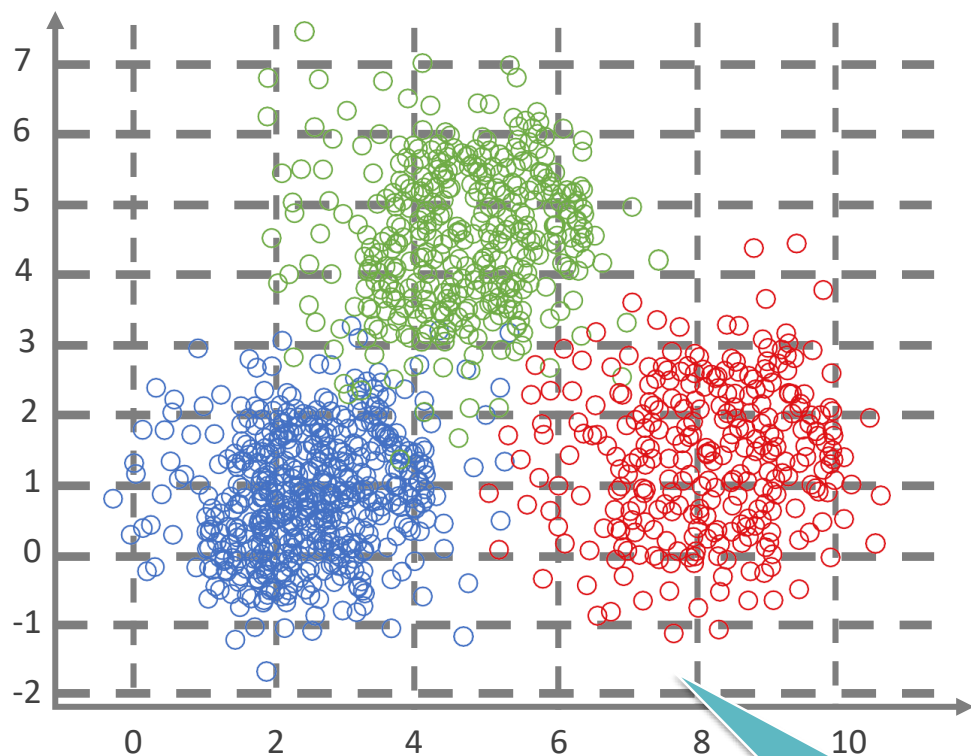


Unsupervised
Learning



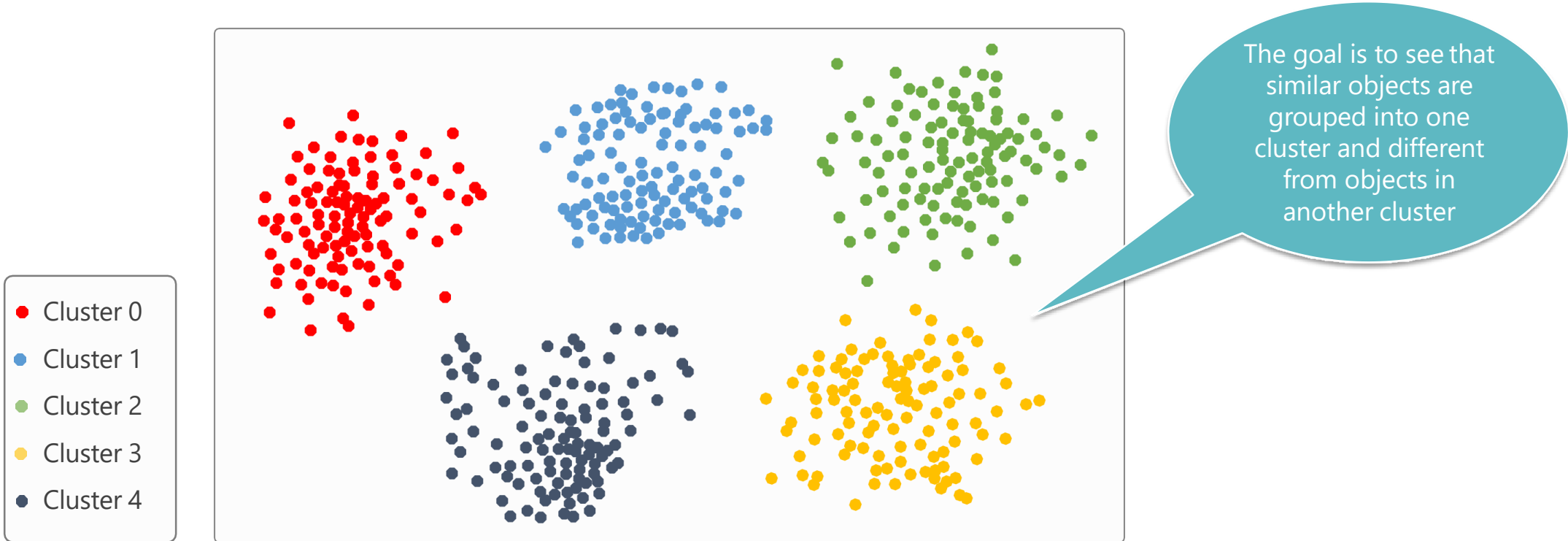
Application of Unsupervised Learning

Unsupervised learning can be used for anomaly detection as well as clustering

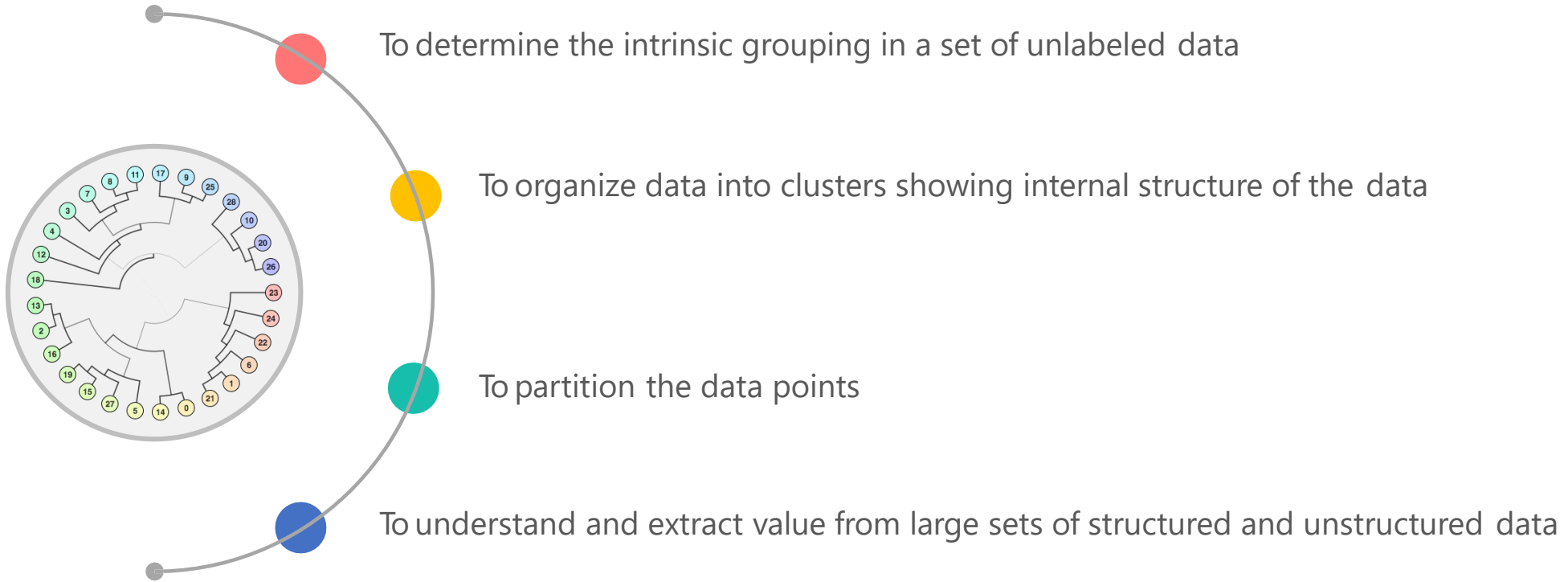


Clustering

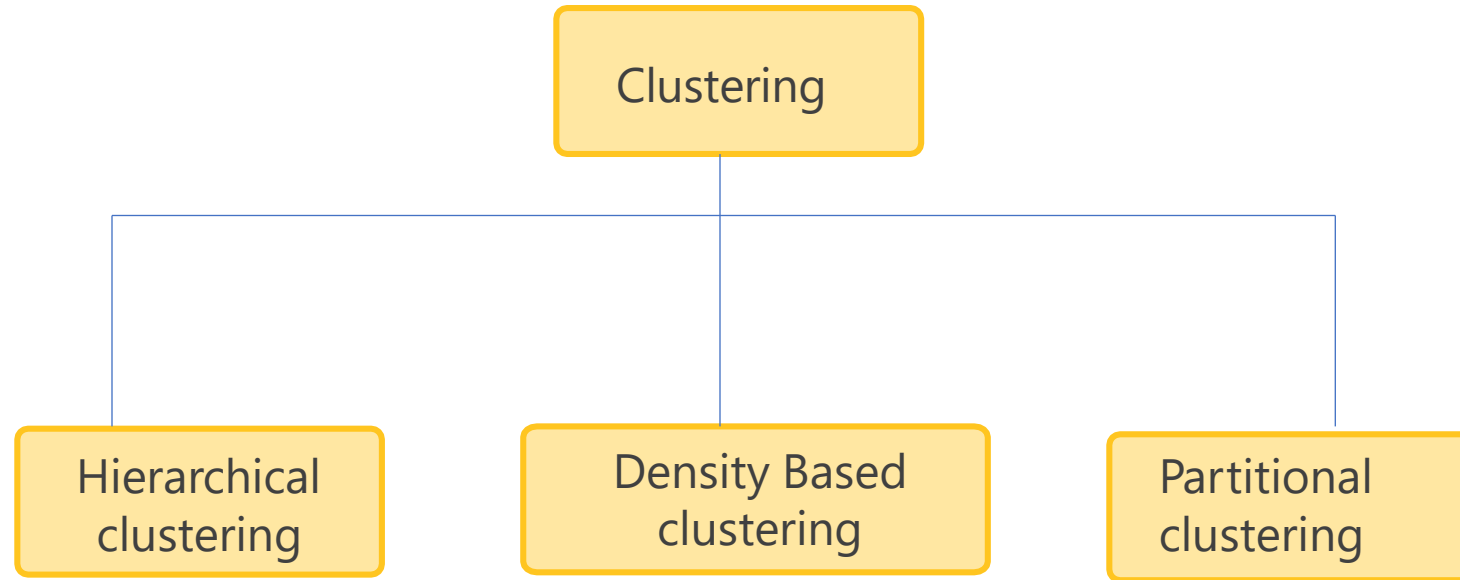
Grouping objects based on the information found in data that describes the objects or their relationship



Need of Clustering



Types of Clustering



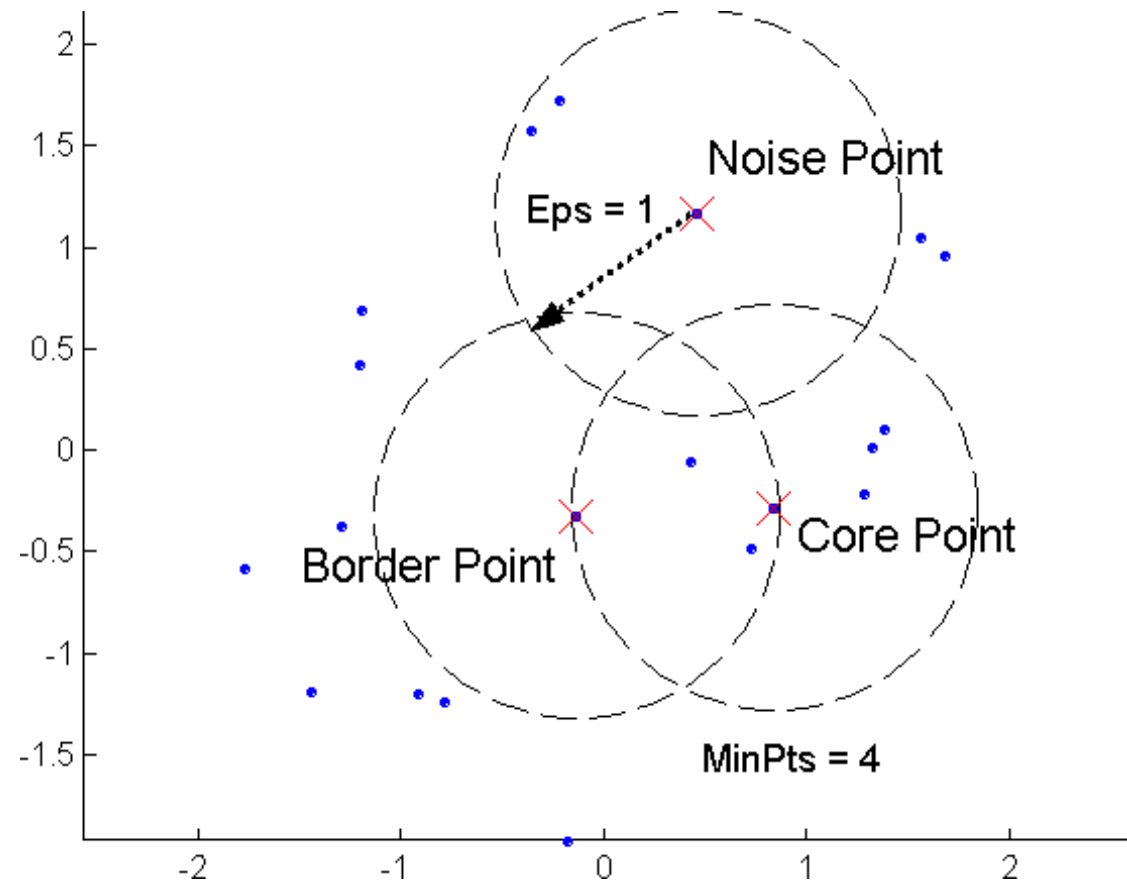
DBSCAN : Density-Based Clustering

- **DBSCAN** is a **Density-Based Clustering** algorithm
- Reminder: In density based clustering we partition points into dense regions separated by not-so-dense regions.
- Important Questions:
 - How do we measure density?
 - What is a dense region?
- **DBSCAN:**
 - **Density at point p**: number of points within a circle of radius **Eps**
 - **Dense Region**: A circle of radius **Eps** that contains at least **MinPts** points

DBSCAN

- Characterization of points
 - A point is a **core point** if it has more than a specified number of points (**MinPts**) within **Eps**
 - These points belong in a **dense region** and are at the **interior** of a cluster
 - A **border point** has fewer than **MinPts** within **Eps**, but is in the neighborhood of a **core** point.
 - A **noise point** is any point that is not a core point or a border point.

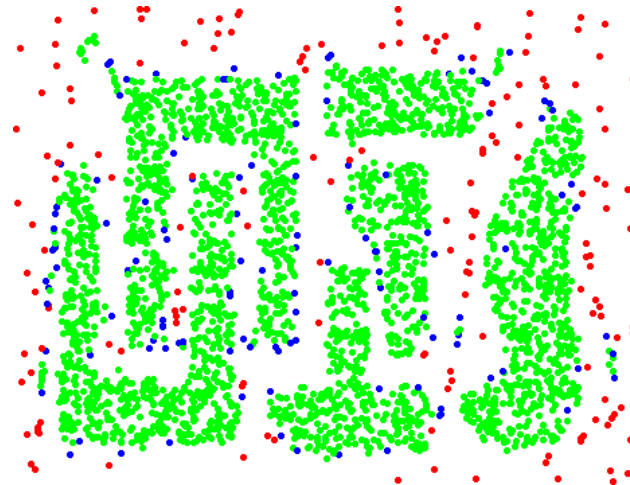
DBSCAN – Core, Border and Noise Points



DBSCAN – Core, Border and Noise Points



Original Points



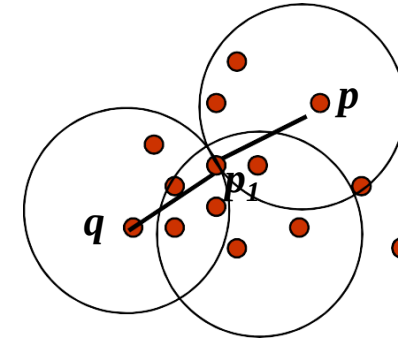
Point types: **core**, **border**
and **noise**

Eps = 10, MinPts = 4

Density Connected Points

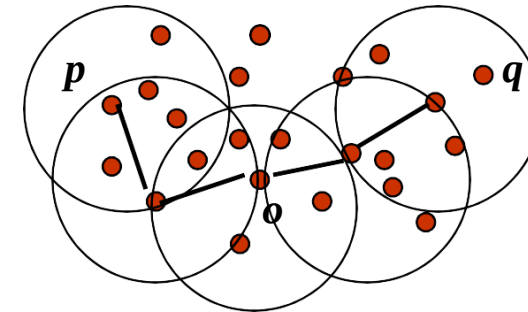
- **Density edge**

- We place an **edge** between two core points **q** and **p** if they are within distance **Eps**.



- **Density-connected**

- A point **p** is **density-connected** to a point **q** if there is a **path of edges** from **p** to **q**



DBSCAN Algorithm

- Label points as **core**, **border** and **noise**
- Eliminate **noise** points
- For every **core** point **p** that has not been assigned to a cluster
 - Create a new cluster with the point **p** and all the points that are **density-connected** to **p**.
- Assign **border** points to the cluster of the closest core point.

Example 1

- Apply the DBSCAN algorithm to the given data points and
- Create the cluster with
- minPts=4 and
- Epsilon(ϵ)=1.9

Data
Points:

P1: (3, 7)

P2: (4, 6)

P3: (5, 5)

P4: (6, 4)

P5: (7, 3)

P6: (6, 2)

P7: (7, 2)

P8: (8, 4)

P9: (3, 3)

P10: (2,6)

P11: (3,5)

P12: (2,4)

Example 1

P1: (3,7)
 P2: (4,6)
 P3: (5,5)
 P4: (6,4)
 P5: (7,3)
 P6: (6,2)
 P7: (7,2)
 P8: (8,4)
 P9: (3,3)
 P10: (2,6)
 P11: (3,5)
 P12: (2,4)

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.88	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

Example 1

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.88	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

Example 1

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

Point	Status	
P1	Noise	Border
P2	Core	
P3	Noise	Border
P4	Noise	Border
P5	Core	
P6	Noise	Border
P7	Noise	Border
P8	Noise	Border
P9	Noise	
P10	Noise	Border
P11	Core	
P12	Noise	Border

Example 1

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

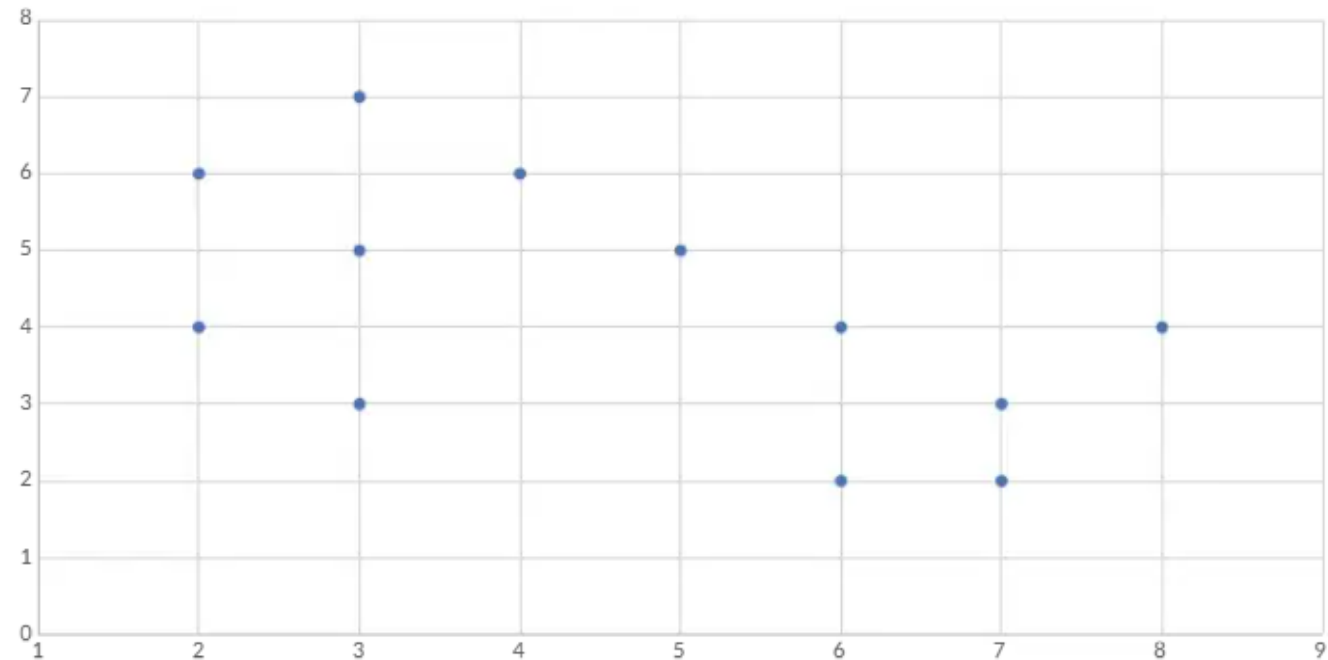
P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11



Evaluation Metrics

- Silhouette Score
- Adjusted Rand Index (ARI)
- Measure clustering quality

Silhouette Score

- Range: -1 to 1
- Near 1 → strong clustering
- Near 0 → overlapping clusters
- Near -1 → incorrect clustering

Silhouette Score Formula

- $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$
- **a(i)** = Average distance between point i and all other points **in the same cluster**
→ measures **cohesion (intra-cluster distance)**
- **b(i)** = Minimum average distance between point i and all points in the **nearest neighboring cluster**

Silhouette Score Interpretation

- Close to +1: well clustered
- Close to 0: overlapping clusters
- Close to -1: misclassified

ARI Formula

$$ARI = \frac{RI - Expected\ RI}{Max\ RI - Expected\ RI}$$

- 1: Perfect match
- 0: Random clustering
- <0: Worse than random

Adjusted Rand Index

- Range: 0 to 1
- >0.9 excellent
- >0.8 good clustering
- <0.5 poor clustering

Example 2

- Perform DBScan on the given problem with $Eps = 2$ and $MinPts = 2$

	X	Y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

Example 2

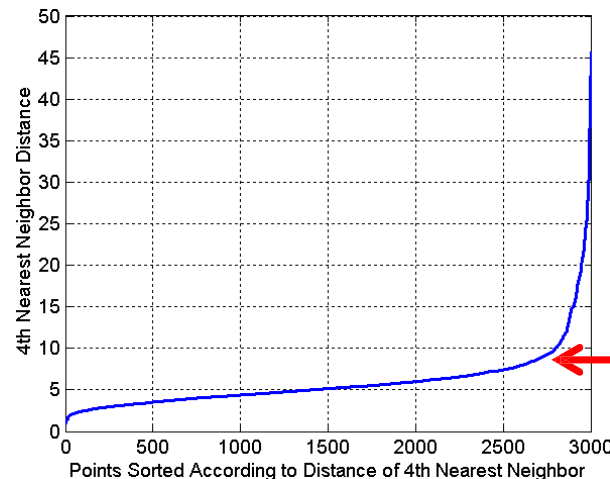
- The foundation of DBSCAN is the distance between every pair of points. Using the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$,

	X	Y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0							
A2	5	0						
A3	8.49	6.08	0					
A4	3.61	4.24	5	0				
A5	7.07	5	1.41	3.61	0			
A6	7.21	4.12	2	4.12	1.41	0		
A7	8.06	3.16	7.28	7.21	6.71	5.39	0	
A8	2.24	4.47	6.4	1.41	5	5.39	7.62	0

DBSCAN – Determining Eps and MinPts

- Idea is that for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance
- Noise points have the k^{th} nearest neighbor at farther distance
- So, plot sorted distance of every point to its k^{th} nearest neighbor
- Find the distance d where there is a “knee” in the curve
 - $\text{Eps} = d, \text{MinPts} = k$

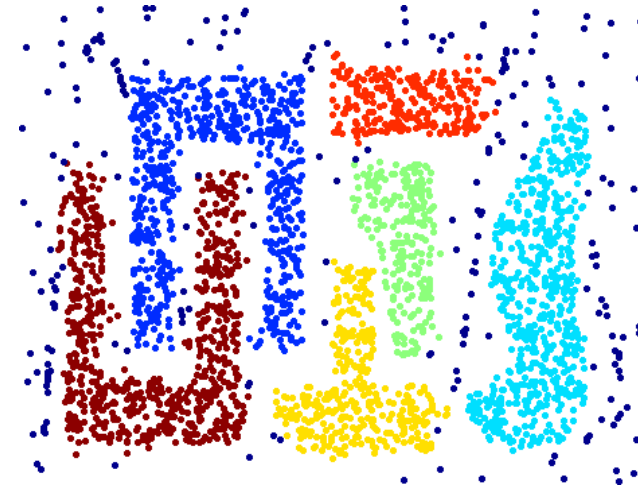


Eps ~ 7-10
MinPts = 4

When DBSCAN Works Well



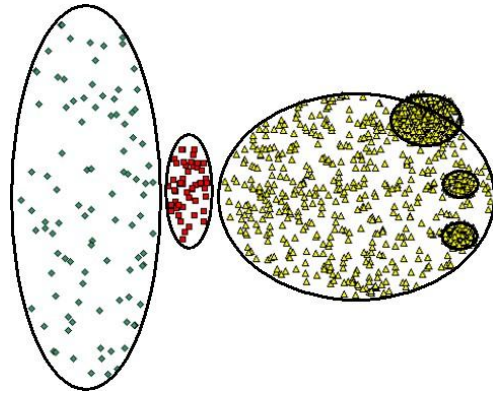
Original Points



Clusters

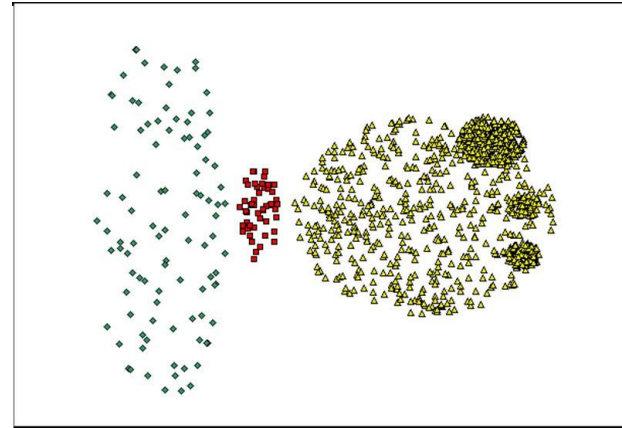
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN not works well

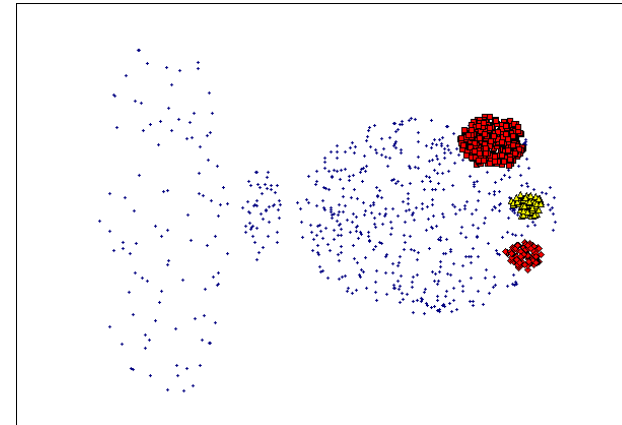


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

DBSCAN Parameter Sensitive

Figure 8. DBScan
 results for DS1 with
 MinPts at 4 and Eps at
 (a) 0.5 and (b) 0.4.

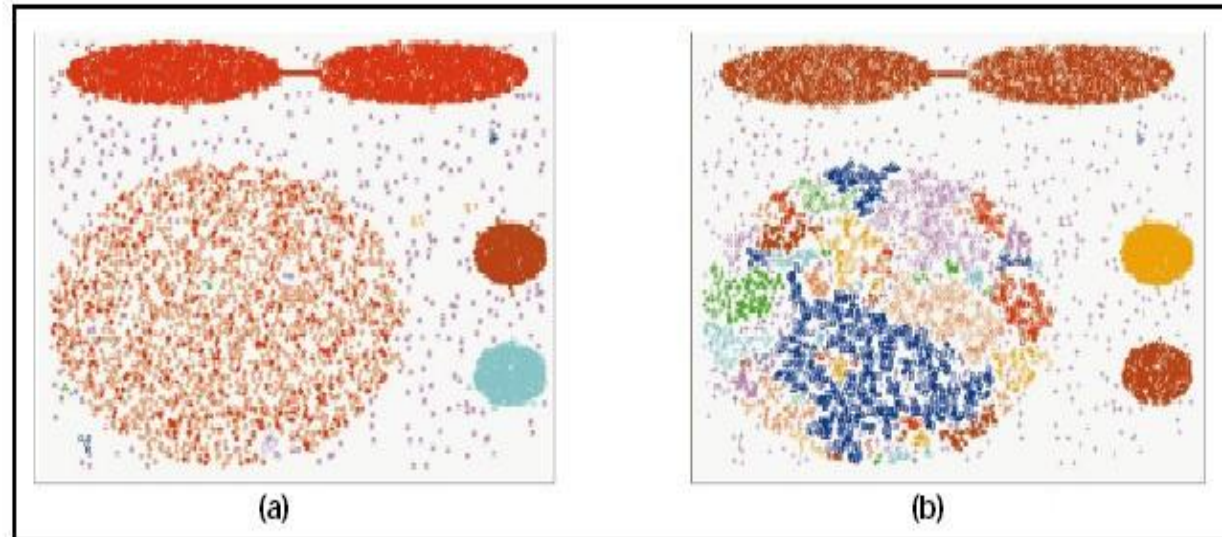
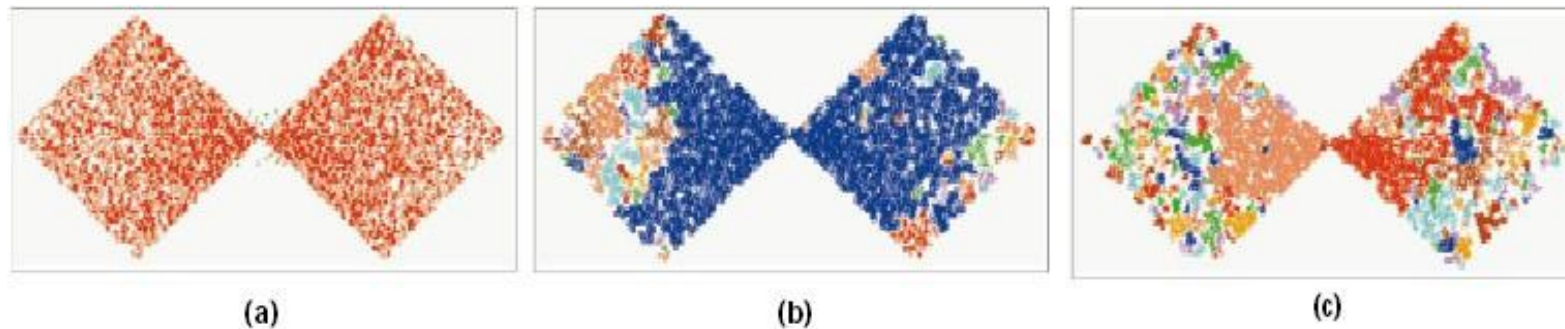


Figure 9. DBScan
 results for DS2 with
 MinPts at 4 and Eps at
 (a) 5.0, (b) 3.5, and
 (c) 3.0.

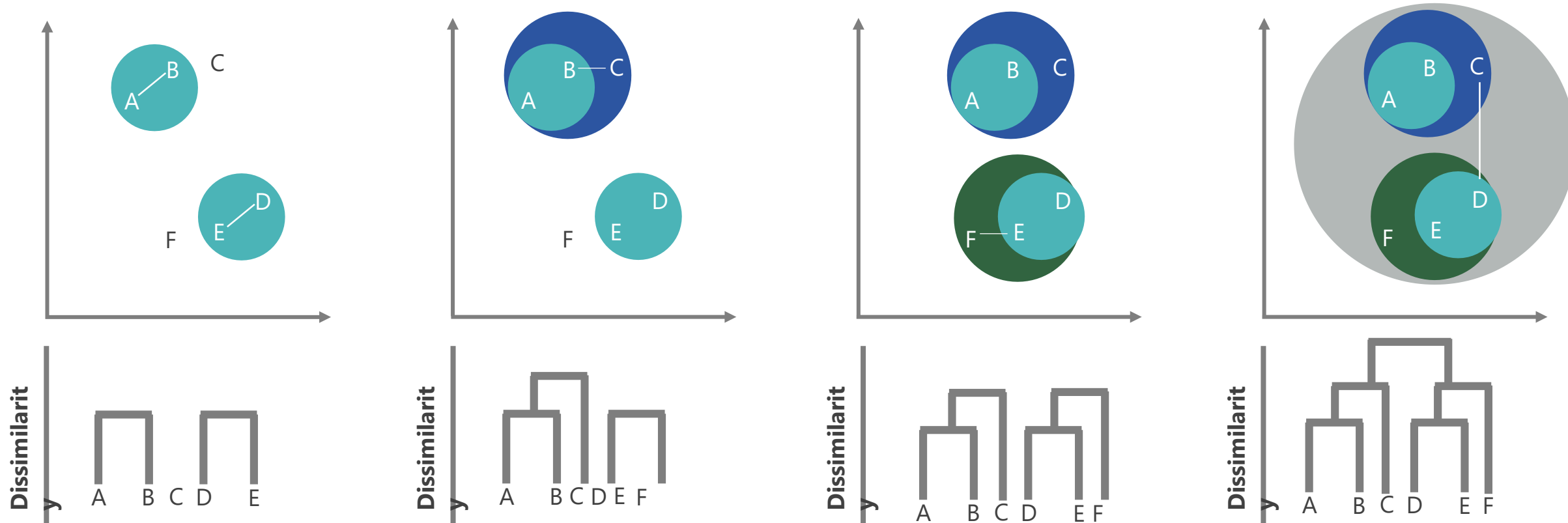


Other algorithms

- **PAM, CLARANS**: Solutions for the **k-medoids** problem
- **BIRCH**: Constructs a **hierarchical tree** that acts a summary of the data, and then clusters the leaves.
- **MST**: Clustering using the **Minimum Spanning Tree**.
- **ROCK**: clustering **categorical data** by neighbor and link analysis
- **LIMBO, COOLCAT**: Clustering **categorical data** using **information theoretic** tools.
- **CURE**: **Hierarchical** algorithm uses different representation of the cluster
- **CHAMELEON**: **Hierarchical** algorithm uses **closeness and interconnectivity** for merging

Hierarchical Clustering

Outputs a hierarchy, a structure that is more informative than the unstructured set of clusters returned by flat clustering



Combine A and B based on similarity
Combine D and E based on similarity

Combination of A and B is combined
with C

Combination of D and E is combined
with F

Final tree contains all clusters
Combined into a single cluster

1

2

3

4

Working: Hierarchical Clustering



Assign each item to its own cluster, such that if you have N number of items, you now have N number of clusters



Find the closest (most similar) pair of clusters and merge them into a single cluster. Now you have one less cluster



Compute distances (similarities) between the new cluster and every old cluster



Repeat steps 2 and 3 until all items are clustered into a single cluster of size N

Distance Measures

Complete - Linkage clustering

- Find the maximum possible distance between points belonging to two different clusters

Single - Linkage Clustering

- Find the minimum possible distance between points belonging to two different clusters

Mean - Linkage Clustering

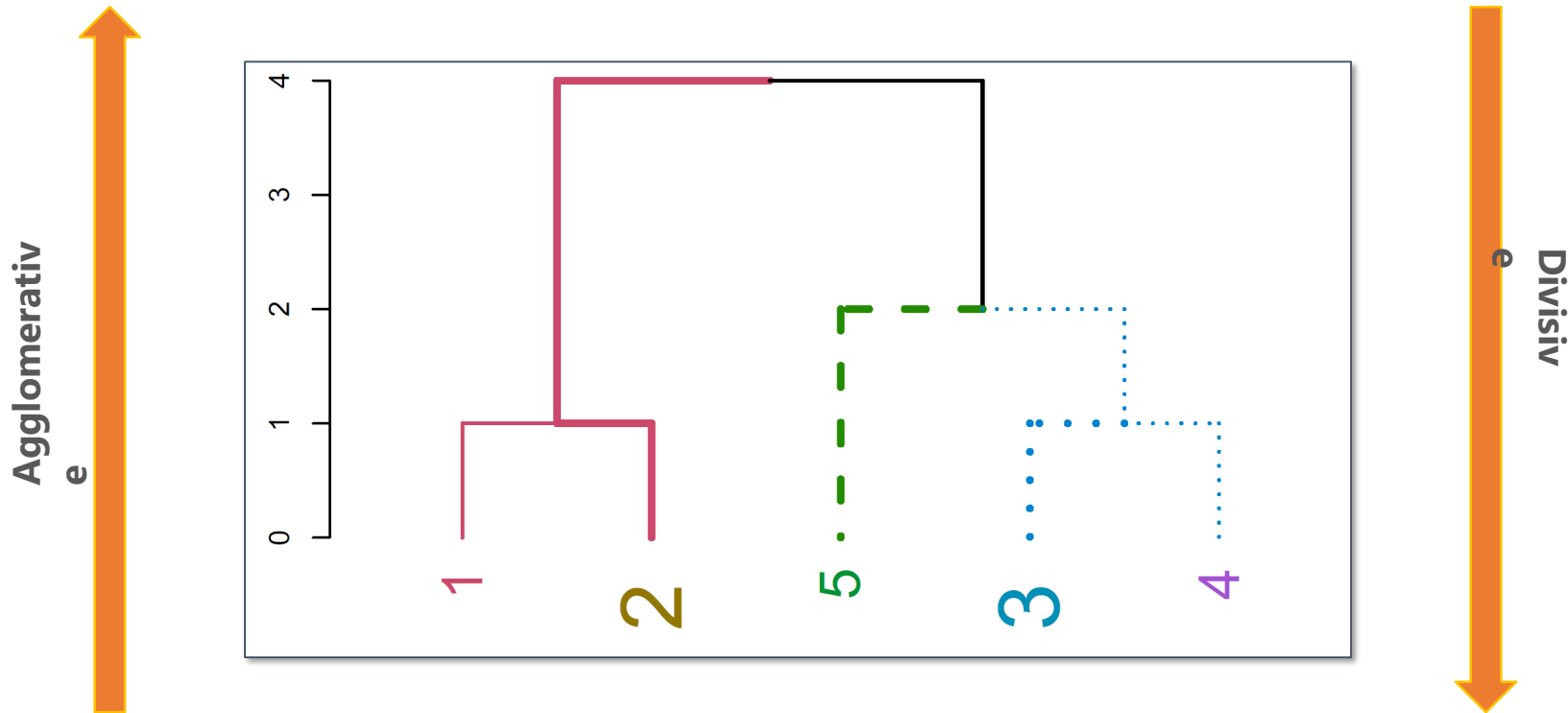
- Find all possible pair-wise distances for points belonging to two different clusters and then calculate the average

Centroid - Linkage Clustering

- Find the centroids of each cluster and calculate the distance between them

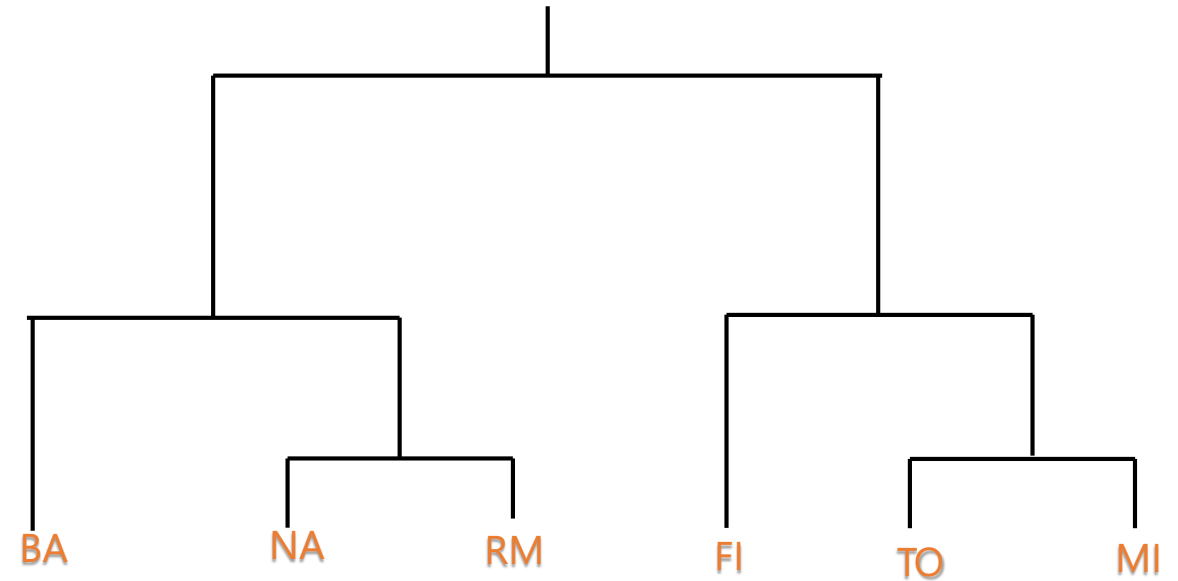
The Dendrogram

Dendrogram ((in Greek, *dendro* means tree and *gramma* means drawing) is a tree diagram frequently used to illustrate the arrangement of the clusters produced by hierarchical clustering.



Hierarchical Clustering: Example

A hierarchical clustering of distances between cities in kilometers



Hierarchical Clustering: Step 1

Create distance matrix of data

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0

Distance between TO and MI

Distance Matrix

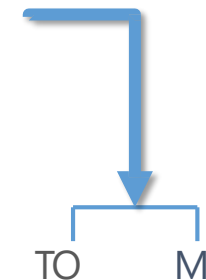
Hierarchical Clustering: Step 2

From the distance matrix, you can see that MI has least distance with TO and they form a cluster together

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0



	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0



As the MI column has lower values than TO column,
MI/TO consists of MI column values

Hierarchical Clustering: Step 3

Repeat clustering until a single cluster is obtained with all the members in it

	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0



	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	468	564	0



Hierarchical Clustering: Step 3 (Contd.)

	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	268	564	0



	BA/(NA/RM)	FI	MI/TO
BA/(NA/RM)	0	268	564
FI	268	0	295
MI/TO	564	295	0



Hierarchical Clustering: Step 3 (Contd.)

	BA/(NA/RM)	FI	MI/TO
BA/(NA/RM)	0	268	564
FI	268	0	295
MI/TO	564	295	0



	BA/(NA/RM)/FI	(MI/TO)
BA/(NA/RM)/FI	0	295
(MI/TO)	295	0

Example

- Construct the dendrogram using Agglomerative Hierarchical Clustering (Single Linkage Method) for the given distance matrix.
- Initially each data point forms a cluster.
- Compute the distance matrix between the clusters.
- Repeat
 - Merge the two closest clusters
 - Update the distance matrix
- Until only a single cluster remains.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Step 1 : {a}, {b}, {c}, {d}, {e}

Step 2 : Distance is given

Step 3: Minimum is 2 ; form c,e as cluster

Example

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Step 3: Updated Distance Matrix after forming cluster {ce}

$d(a, \{ce\})$, $d(d, \{ce\})$

	a	b	d	ce
a	0	9	6	3
b	9	0	5	6
d	6	5	0	8
ce	3	6	8	0

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b}, {c,e}, {d}

Example

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

- Step 3: Minimum distance = 3 $\rightarrow$ Merge (a, {ce}) $\rightarrow$ New cluster = {ace}
- Updated Distance Matrix after forming cluster {ace}
- $d(b, \{ace\})$, $d(d, \{ace\})$

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b}, {c, e}, {d}

{a, c, e}, {b}, {d}

	ace	b	d
ace	0	6	6
b	6	0	5
d	6	5	0

	ace	b	d
ace	0		
b		0	
d			0

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

- Step 3: Minimum distance = 5 $\rightarrow$ Merge (b, d) $\rightarrow$ New cluster = {bd}

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b}, {c,e}, {d}

{a,c,e}, {b},{d}

{a,c,e}, {b,d}

Example

- Complete Linkage and Average Linkage Solved

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Advantages

- Easy to understand
- No need to choose number of clusters initially
- Dendrogram gives visual representation

Disadvantages

- Computationally expensive for large datasets
- Sensitive to noise and outliers
- Once merged, clusters cannot be split

Applications

- Gene expression analysis
- Document clustering
- Customer segmentation
- Image segmentation



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Applications

Refer Machine Learning book by Anuradha